HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented automated test scripts for implantable medical device components, reducing manual testing effort by **40%**.
- Performed root cause analysis on electronic component failures for implantable devices, identifying critical defects in **solder joint integrity** that led to a **15% reduction** in early-stage product failures.
- Collaborated with software developers to integrate new testing software into existing manufacturing systems, improving test data capture efficiency by **22%**.
- Contributed to the validation of new test fixtures and equipment, ensuring compliance with medical device standards and preparing technical documentation for regulatory submissions.
- Analyzed manufacturing test data using **SQL** and **Python** to identify trends and areas for process improvement, presenting findings to cross-functional teams.
- Supported the implementation of process improvements focused on enhancing product quality and test system efficiency, contributing to a **10% decrease** in production fallout.
- Managed the change control process for manufacturing system updates, coordinating User Acceptance Testing and providing post-go-live support for interconnected systems.

SKILLS

Engineering & Testing – Test Development, Failure Analysis, Process Improvement, Lean Manufacturing, Validation, Root Cause Analysis, Quality Assurance

Programming & Scripting – Python, SQL, LabVIEW, C, C++

Data Analysis & BI – Data Analysis, Technical Reporting, Power BI, Tableau, DAX, Power Query

Software & Tools – Jira, Confluence, Git, Jenkins, Docker, Linux, Splunk, Dynatrace

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch)

Domain Knowledge – Medical Devices, Electronics Manufacturing, Implantable Devices, Regulatory Compliance

PROJECTS

Automated Test System for Medical Device Components

- Developed automated test scripts using **Python** and **LabVIEW** to verify functionality and performance of critical electronic components in implantable devices.
- Integrated test scripts with internal component tracking systems, enabling real-time data logging and facilitating faster root cause analysis for failures.
- Prepared comprehensive technical reports detailing test results and failure modes, supporting the Quality Assurance team's validation efforts.

Manufacturing Process Data Analysis and Improvement

- Extracted and analyzed manufacturing process data using **SQL** and **pandas** to identify key drivers of yield loss and product defects.
- Built interactive **Power BI** dashboards to visualize process performance metrics and failure trends for manufacturing and engineering teams.
- Collaborated with operations teams to implement data-driven process adjustments, resulting in a measurable improvement in test yields.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)